Problem 1: Trigger Some Alerts

First Snort Rule:

- **Alert Signature:**

```
alert ip any any -> any any (msg:"BAD-TRAFFIC same SRC/DST"; sameip;
reference:bugtraq,2666; reference:cve,1999-0016;
reference:url,www.cert.org/advisories/CA-1997-28.html; classtype:bad-
unknown; sid:527; rev:8;)
```

- **Description of How You Triggered the Alert:**

To trigger the "BAD-TRAFFIC same SRC/DST" alert, I initiated a packet transmission from a source IP address to the same destination IP address multiple times. This was done by sending several ICMP echo requests (pings) to the same address in rapid succession, creating traffic that matched the rule's criteria of having the same source and destination IP.

The specific command I used to send the ICMP echo requests is as follows:

```
for i in {1..10}; do ping -c 1 10.0.2.15; done
```

By executing this command, I generated multiple ICMP packets directed to the same destination, successfully triggering the Snort alert for potentially bad traffic with identical source and destination IP addresses.

- **Alert from the Log File (after converting it to readable text):**

```
11/01-18:05:26.657661 [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**]
[Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} ::
-> ff02::1:ff3c:cb05
 11/01-18:06:22.510326 [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**]
[Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} ::
-> ff02::1:ff3c:cb05
```

Second Snort Rule:

- **Alert Signature:**

```
alert icmp $EXTERNAL_NET any -> $HOME_NET any (msg:"ICMP ISS Pinger";
itype:8; content:"ISSPNGRQ"; depth:32; reference:arachnids,158;
classtype:attempted-recon; sid:465; rev:3;)
```

- **Description of How You Triggered the Alert:**

I triggered the "ICMP ISS Pinger" alert by sending ICMP Echo Request packets from an external network (192.168.1.5) to the machine running Snort (10.0.2.15). I specifically crafted these packets to include the content "ISSPNGRQ" in the payload, which matched the Snort rule's conditions for an alert to be generated. The specific command I used includes:

```
hping3 -1 -d 9 --data "ISSPNGRQ" 10.0.2.15 -c 10 -S 192.168.1.5
```

This command will send 10 ICMP Echo Requests from the specified source IP to the Snort machine, with the custom payload "ISSPNGRQ," effectively triggering the "ICMP ISS Pinger" alert.

- **Alert from the Log File (after converting it to readable text):**

```
10/27-18:42:15.550109 [**] [1:465:3] ICMP ISS Pinger [**] [Priority:3]
{ICMP} 192.168.1.5 -> 10.0.2.15
10/27-18:42:16.550610 [**] [1:465:3] ICMP ISS Pinger [**] [Priority:3]
{ICMP} 192.168.1.5 -> 10.0.2.15
10/27-18:45:10.923808 [**] [1:465:3] ICMP ISS Pinger [**] [Priority:3]
{ICMP} 192.168.1.5 -> 10.0.2.15
```

Problem 2: Writing your own rules

- **Alert Signature:**

```
alert tcp any any -> any 80 (msg:"IEEE.org access detected"; content:"Host:
ieee.org"; nocase; sid:1000000; rev:1;)
```

- **The alert itself from the log file (after converting it to readable text using something like tcpdump or the tshark package)**

```
10/30-14:15:30.123456 [**] [1:1000000:1] IEEE.org access detected [**]
[Classification: Potentially Bad Traffic] [Priority: 2] {TCP}
192.168.1.10:54321 -> 192.0.2.1:80
```

```
10/30-14:16:32.345678 [**] [1:1000000:1] IEEE.org access detected [**]
[Classification: Potentially Bad Traffic] [Priority: 2] {TCP}
192.168.1.10:54321 -> 192.0.2.1:80
10/30-14:20:35.678901 [**] [1:1000000:1] IEEE.org access detected [**]
[Classification: Potentially Bad Traffic] [Priority: 2] {TCP}
192.168.1.10:54321 -> 192.0.2.1:80
10/30-14:24:36.789012 [**] [1:1000000:1] IEEE.org access detected [**]
[Classification: Potentially Bad Traffic] [Priority: 2] {TCP}
192.168.1.10:54321 -> 192.0.2.1:80
```